Cambridge Assessment International Education

# Cambridge O Level

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

*3764988855*

**ADDITIONAL MATHEMATICS** **4037/12**

Paper 1 **May/June 2023**

**2 hours**

You must answer on the question paper.

No additional materials are needed.

**INSTRUCTIONS**

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.

**INFORMATION**

- The total mark for this paper is 80.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **16** pages. Any blank pages are indicated.


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## *Mathematical Formulae*

### 1. ALGEBRA

*Quadratic Equation*

For the equation $ax^2 + bx + c = 0$,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

*Binomial Theorem*

$$(a+b)^n = a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \ldots + \binom{n}{r}a^{n-r}b^r + \ldots + b^n$$

where $n$ is a positive integer and $\binom{n}{r} = \frac{n!}{(n-r)!r!}$

*Arithmetic series*

$$u_n = a + (n-1)d$$

$$S_n = \frac{1}{2}n(a+l) = \frac{1}{2}n\{2a + (n-1)d\}$$

*Geometric series*

$$u_n = ar^{n-1}$$

$$S_n = \frac{a(1-r^n)}{1-r} \quad (r \neq 1)$$

$$S_\infty = \frac{a}{1-r} \quad (|r| < 1)$$

### 2. TRIGONOMETRY

*Identities*

$$\sin^2 A + \cos^2 A = 1$$
$$\sec^2 A = 1 + \tan^2 A$$
$$\operatorname{cosec}^2 A = 1 + \cot^2 A$$

*Formulae for $\Delta ABC$*

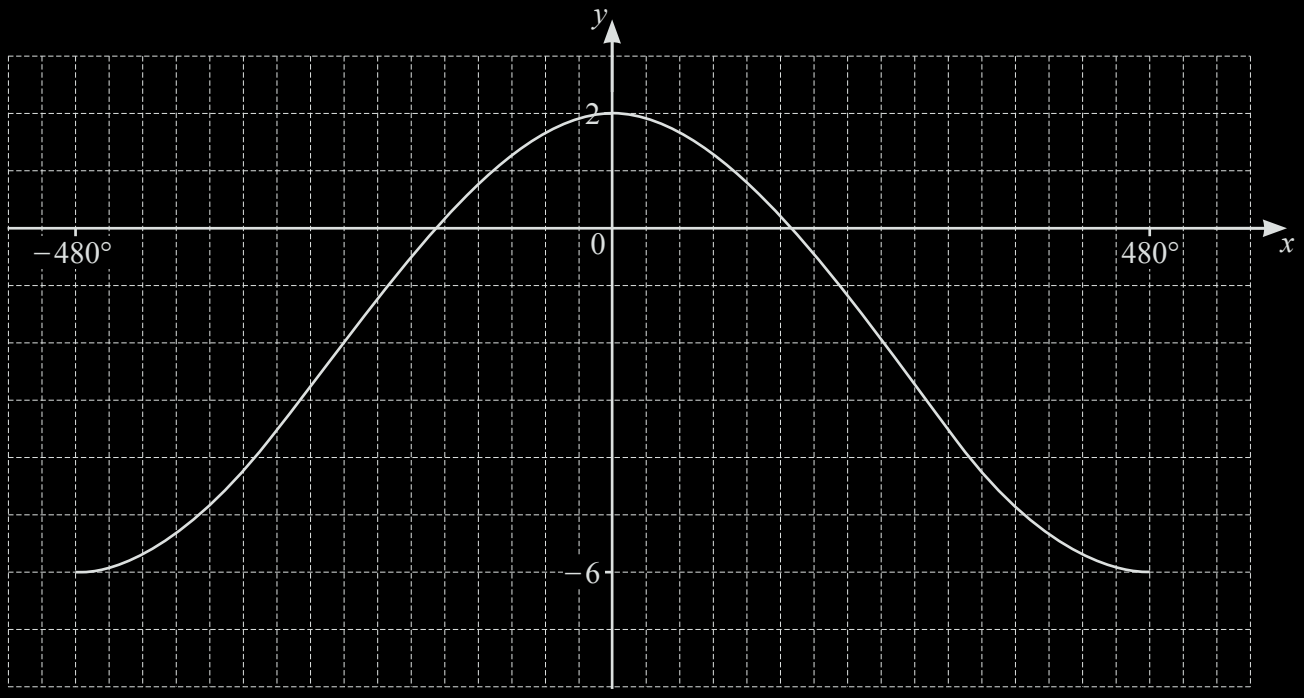
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc\cos A$$

$$\Delta = \frac{1}{2}bc\sin A$$

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**1** The diagram shows the graph of $y = a\cos bx + c$. Find the values of the constants $a$, $b$ and $c$. [3]

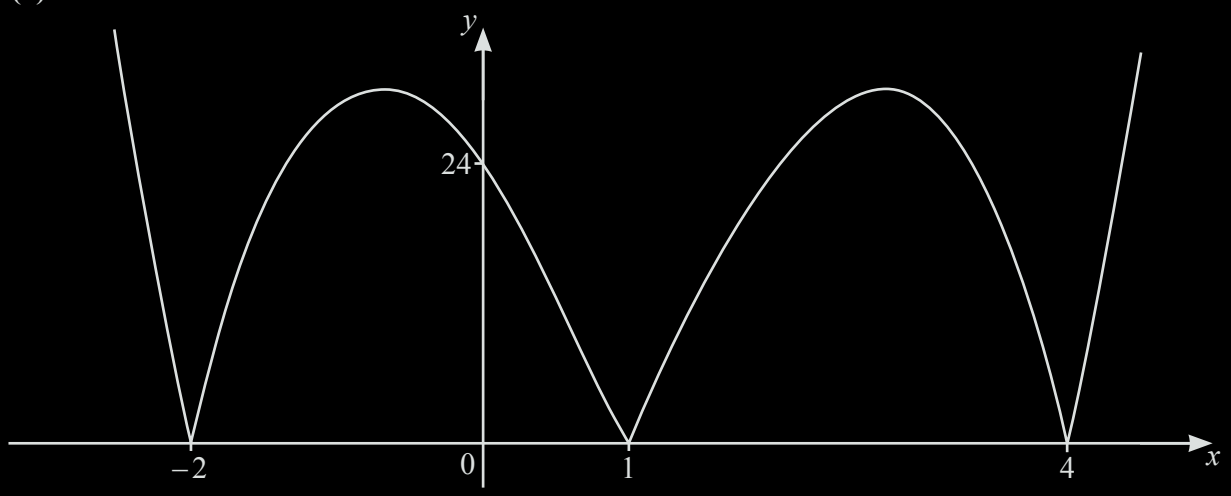


**2 DO NOT USE A CALCULATOR IN THIS QUESTION.**

Solve the equation $\left(2+\sqrt{5}\right)x^2 = 4x+3\left(2-\sqrt{5}\right)$, giving your answers in the form $a+b\sqrt{5}$ where $a$ and $b$ are integers. [5]

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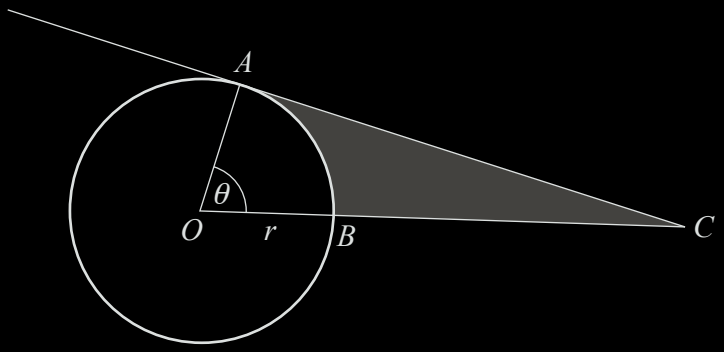
**3** **(a)**




The diagram shows the graph of $y = |f(x)|$, where $f(x)$ is a cubic polynomial. Find, in factorised form, the possible expressions for $f(x)$. [3]

**(b)** Solve the inequality $|5x - 2| \leqslant |4x + 1|$. [4]

**4** In this question all lengths are in centimetres and all angles are in radians.




The diagram shows a circle with centre $O$ and radius $r$. The points $A$ and $B$ lie on the circumference of the circle such that the angle $AOB$ is $\theta$ and the length of the minor arc $AB$ is 12. The area of the minor sector $AOB$ is $57.6\,\text{cm}^2$. The point $C$ lies on the tangent to the circle at $A$ such that $OBC$ is a straight line.

**(a)** Find the values of $r$ and $\theta$. [4]

**(b)** Find the area of the shaded region. Give your answer correct to 1 decimal place. [3]

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**5** **(a)** Find the exact solutions of the equation $6p^{\frac{1}{3}} - 5p^{-\frac{1}{3}} - 13 = 0$. [4]

**(b)** Solve the equation $2\lg(2x+5) - \lg(x+2) = 1$, giving your answers in exact form. [6]

**6** **(a)** Given that $\cot^2\theta = \frac{1}{y+2}$ and $\sec\theta = x-4$, find $y$ in terms of $x$. [2]

**(b)** Solve the equation $\sqrt{3}\,\text{cosec}\left(2\phi + \frac{3\pi}{4}\right) = 2$, for $-\pi < \phi < \pi$, giving your answers in terms of $\pi$. [5]

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7 **(a)** Find the number of ways in which 14 people can be put into 4 groups containing 2, 3, 4 and 5 people. [3]

**(b)** 6-digit numbers are to be formed using the digits 0, 1, 2, 3, 4, 5, 6, 7, 8, 9. Each digit may be used only once in any 6-digit number. A 6-digit number must not start with 0. Find how many 6-digit numbers can be formed if

**(i)** there are no further restrictions [1]

**(ii)** the 6-digit number is divisible by 10 [1]

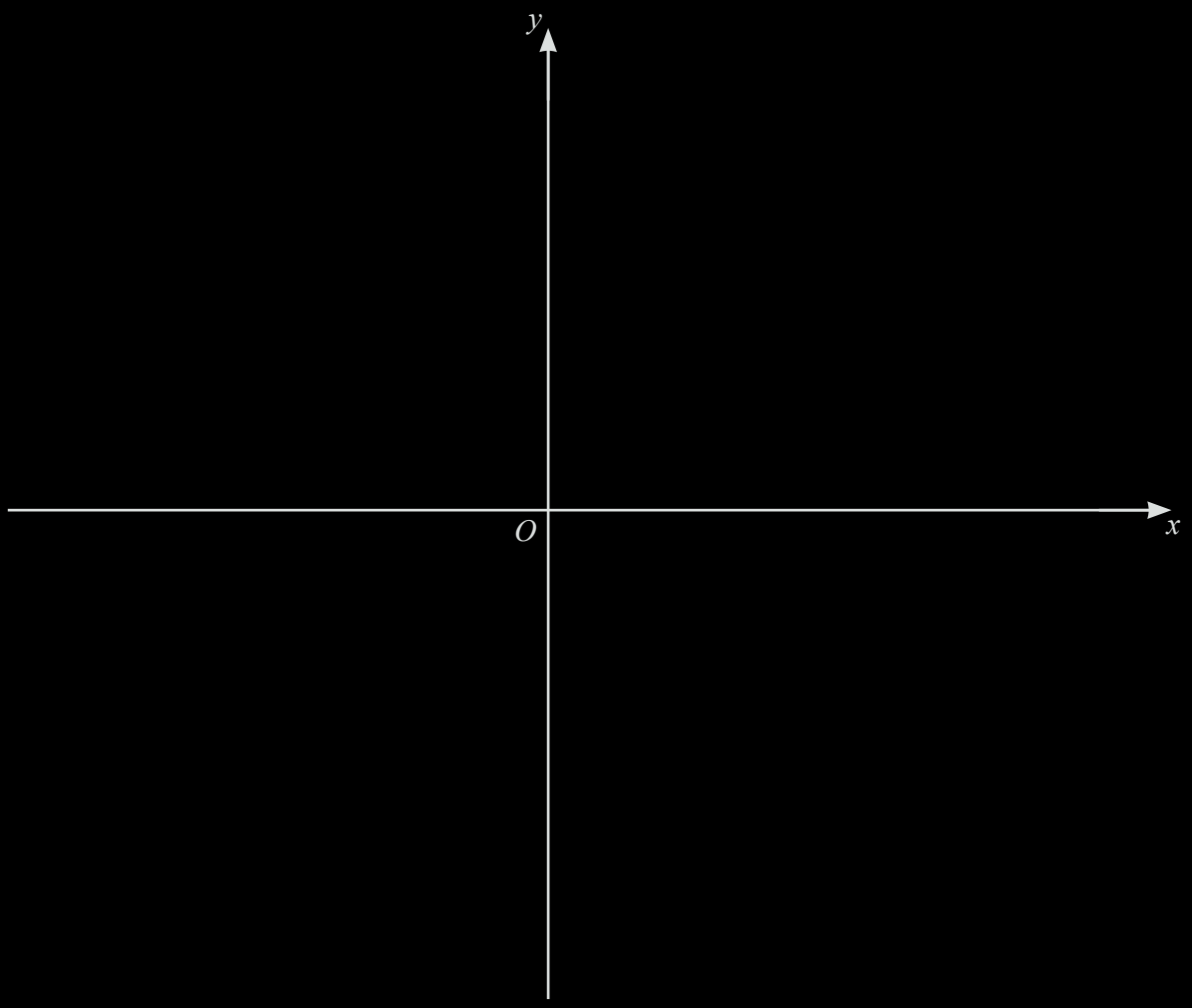
**(iii)** the 6-digit number is greater than 500 000 and even. [3]

**8** It is given that $\mathrm{f}(x) = 2\ln(3x-4)$ for $x > a$.

**(a)** Write down the least possible value of $a$. [1]

**(b)** Write down the range of f. [1]

**(c)** It is given that the equation $\mathrm{f}(x) = \mathrm{f}^{-1}(x)$ has two solutions. (You do not need to solve this equation). Using your answer to **part (a)**, sketch the graphs of $y = \mathrm{f}(x)$ and $y = \mathrm{f}^{-1}(x)$ on the axes below, stating the coordinates of the points where the graphs meet the axes. [4]



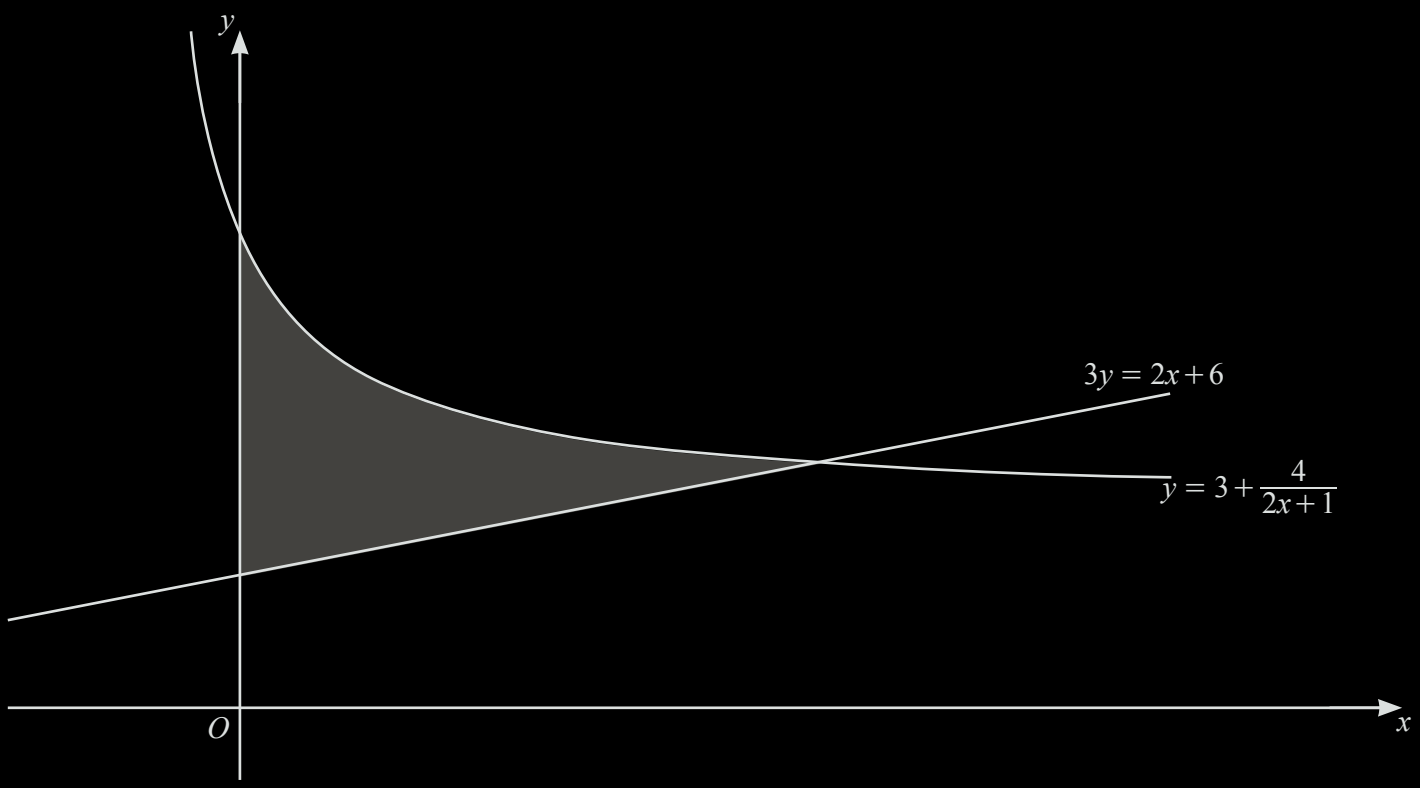

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It is given that $g(x) = 2x - 3$ for $x \geqslant 3$.

**(d) (i)** Find an expression for $g(g(x))$. [1]

**(ii)** Hence solve the equation $fg(g(x)) = 4$ giving your answer in exact form. [3]

**9**




The diagram shows part of the curve $y = 3 + \frac{4}{2x+1}$ and the straight line $3y = 2x + 6$. Find the area of the shaded region, giving your answer in exact form. [10]

Continuation of working space for Question 9.

**10** **(a)** The first three terms of an arithmetic progression are $(2x+1)$, $4(2x+1)$ and $7(2x+1)$, where $x \neq -\frac{1}{2}$.

**(i)** Show that the sum to $n$ terms can be written in the form $\frac{n}{2}(2x+1)(An+B)$, where $A$ and $B$ are integers to be found. [2]

**(ii)** Given that the sum to $n$ terms is $(54n+37)(2x+1)$, find the value of $n$. [2]

**(iii)** Given also that the sum to $n$ terms in **part (ii)** is equal to 1017.5, find the value of $x$. [2]

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**(b)** The first three terms of a geometric progression are $(2y+1)$, $3(2y+1)^2$ and $9(2y+1)^3$, where $y \neq -\frac{1}{2}$.

Given that the $n$th term of the progression is equal to 4 times the $(n+2)$th term, find the possible values of $y$, giving your answers as fractions. [4]

**(c)** The first three terms of a different geometric progression are $\sin\theta$, $2\sin^3\theta$ and $4\sin^5\theta$, for $0 < \theta < \frac{\pi}{2}$. Find the values of $\theta$ for which the progression has a sum to infinity. [3]

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